Rafael Martínez Estévez

Senior Software Engineer · Technical Lead · Open Source Advocate

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SUMMARY

Results-driven Software Engineer and Technical Lead with 15+ years of experience designing, building, and deploying scalable, open-source data platforms. Expertise in full-stack and backend development using Python, FastAPI, Angular, and graph databases (Neo4j). Proven track record of leading cross-functional teams to deliver complex knowledge graph, LLM-integrated, and metadata interoperability systems. Strong advocate for modern DevOps practices, CI/CD, and open-source collaboration, with a long-standing research trajectory in applied Machine Learning and AI.

EXPERIENCE

Assistant Professor & Technical Lead
2016 – Present

Universidad de Pinar del Río (UPR) — Pinar del Río, Cuba

Lead the architecture and development of institutional research platforms, transitioning legacy systems to modern, containerized microservices.

- Administer and optimize high-availability institutional repositories and scientific journal platforms (Invenio, DSpace, OJS), ensuring robust data interoperability via OAI-PMH and REST APIs.
- Mentor junior developers and coordinate agile development cycles for university-wide software initiatives, fostering a culture of code review and continuous integration.
- Teach *Introduction to Artificial Intelligence*, integrating modern machine learning methodologies and project-based software development practices.

Python · FastAPI · Angular · Docker · Neo4j

Instructor & Full-Stack Developer
2010 – 2015

Rafael María de Mendive University of Pedagogical Sciences — Pinar del Río, Cuba

Designed and delivered technical courses in Programming, Computer Architecture, and Databases.

- Developed and deployed custom full-stack web applications for local institutions, managing the entire software development lifecycle from requirements gathering to production deployment.

Python · JavaScript · MySQL · Flask

Systems Developer & Adjunct Instructor
2007 – 2009

Ministry of Economy and Planning / CUJAE — Havana, Cuba

- Built and maintained enterprise-level information systems to support national economic data analysis and planning.
- Taught *Introduction to Programming* and *Introduction to Computer Science* at the University Headquarters of the Plaza de la Revolución Municipality.

Java · SQL · Linux

PROJECTS & OPEN SOURCE

Iroko-CRIS & Sceiba Platform
2022 – Present

Project Leader, Analyst and Developer

Architected a modern Current Research Information System (CRIS) to aggregate, monitor, and evaluate national scientific publications.

- Designed a hybrid information retrieval system combining graph databases (Neo4j) and full-text search (Typesense/Elasticsearch) via a FastAPI backend and Angular frontend.

- Established data models based on semantic web ontologies to ensure international metadata interoperability and seamless data exchange.

  `Python · FastAPI · Neo4j · Angular · Typesense`

UPR-K Platform 2024 – Present
Project Leader, Analyst and Developer

Spearheading the development of a next-generation library platform leveraging Knowledge Graphs (KG) and Large Language Models (LLM) to enhance user interaction with institutional data and reimagine academic library services.

`Python · LLM · Knowledge Graphs · FastAPI`

Tocororo Open Source Organization 2018 – Present
Core Maintainer & Coordinator

Manage and actively contribute to multiple open-source repositories (e.g., `iroko-cris`, `invenio-cli`, `rero-ils`), focusing on Python and TypeScript development.

- Drive community collaboration, code quality, and architectural decisions across distributed development teams.

  `Python · TypeScript · GitHub Actions`

SKILLS

Languages Python, TypeScript, JavaScript, Java, Cypher, SQL
Backend & Frameworks FastAPI, Flask, Angular, Node.js, RESTful APIs, Semantic Web
Databases & Search Neo4j, Elasticsearch, Typesense, MongoDB, MySQL
DevOps & Infrastructure Docker, Podman, LXD, Proxmox, Ansible, CI/CD
Machine Learning & AI Large Language Models (LLM), Multilingual BERT, Fine-Tuning, Pattern Recognition

EDUCATION

Bachelor of Computer Science 2002 – 2007
Faculty of Mathematics and Computing, University of Havana Cuba

- Bachelor's Thesis: *"Vocal Segmentation for Speaker Recognition using Machine Learning Algorithms."* Early applied research in AI, focusing on pattern recognition, signal processing, and the implementation of ML models for audio classification.

PUBLICATIONS & PRESENTATIONS

The Impact of the Dataset on Fine-Tuning Multilingual BERT 2026
European Conference on Artificial Intelligence (ECAI 2026) Abreu-Hernández, E., Martínez-Estévez, R., Savón-Berenguer, A., & Moraru, S.A.

Construyendo el bien común: el rol del código abierto... 2025
XI International Workshop on Free and Open Source Software (UCIencia) Martínez Estévez, R.

Sceiba, as a national platform for research information 2023
WebOLInFER 2023, Universidad de Ciencias Informáticas, Cuba Martínez Estévez, R.

ELINFINTER: model for data interoperability in HEIs 2022
Metadata and Semantic Research Conference (MTSR) Leiva Mederos, A., Martínez Estévez, R., et al.

Identificacion de organizaciones en la Plataforma Sceiba 2022
Hatun Tinkuy 2022, Concytec, Perú. Martínez Estévez, R., & Lorenzo Cabezas, R.

On metadata quality in Sceiba, a platform for quality control and monitoring of Cuban scientific publications 2021
Arencibia, E., Martínez Estévez, R., et al.

Viabilidad de Python en la enseñanza de la programación 2014
Mendive. Revista de Educación, 12(2), 179-186 Martínez Estévez, R., Pereira Rosa, M., & González Fernández, R.

> CERTIFICATIONS & TRAINING

Linux Essentials 2023
Linux Professional Institute (LPI)

Flemish Research Repository Training 2019
University of Antwerp, Belgium

Big Data Analytics 2018
VLIR Project, University of Pinar del Río, Cuba

Semantic Web and Linked Data 2017
Marta Abreu de las Villas Central University, Cuba